

TOPIC NAME: Gradient boost

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Gradient boosting for regression

Height	Favourite color	Gender	Weight (kg)	Predicted weight
1.6	Blue	Male	88	71.2
1.6	Green	Female	76	71.2
1.5	Blue	Female	56	71.2
1.8	Red	Male	73	71.2
1.5	Green	Male	77	71.2
1.4	Blue	Female	57	71.2

Step 1: Calculate average of the target column.

$$\text{Initial prediction / guess} = \frac{88 + 76 + 56 + 73 + 77 + 57}{6}$$

$$= 71.2$$

Step 2: calculate residual

pseudo residual = actual / observed weight - predicted weight

Weight	Predicted	residual	→ error
88	71.2	16.8	
76	71.2	4.8	
56	71.2	-15.2	
73	71.2	1.8	
77	71.2	5.8	
57	71.2	-14.2	

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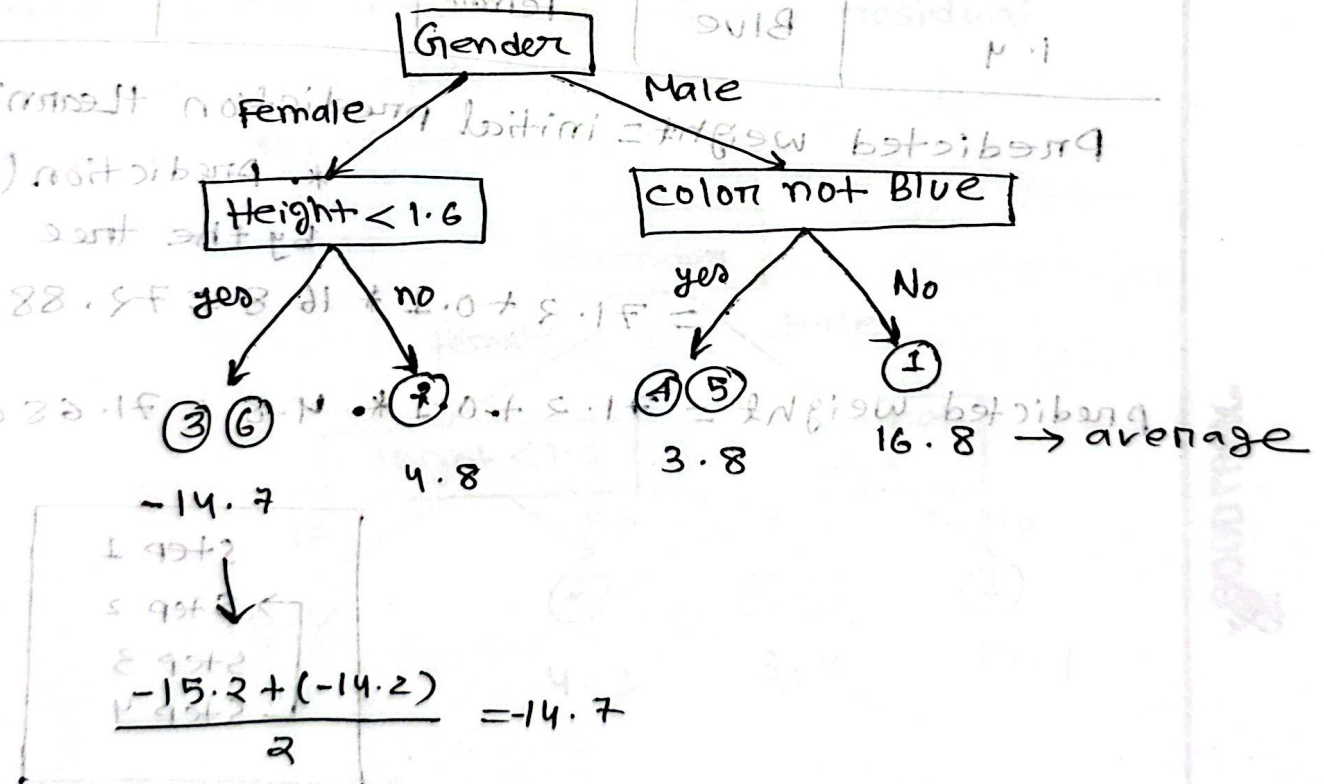
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Step 3: Build a tree to predict the residual

	Height	color	gender	Residual	new target
①	1.6	Blue	Male	16.8	
②	1.6	Green	Female	4.8	
③	1.5	Blue	Female	-15.2	
④	1.8	Red	Male	1.8	
⑤	1.5	Green	Male	5.8	
⑥	1.4	Blue	Female	-14.2	

Regression Tree



Gradient boosting is a tree where leaf nodes have a limit on the number of observations.

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Step 3: Build a tree to predict the target value
limit = No of leaf

Step 4: Update the predicted value of the target.

Height	Color	Gender	Weight	updated predicted weight
1.6	Blue	Male	88	72.88
1.6	Green	Female	76	71.68
1.5	Blue	Female	56	69.7
1.8	Red	Male	73	71.6
1.5	Green	Male	77	71.6
1.4	Blue	Female	57	69.7

Predicted weight = initial prediction + Learning rate

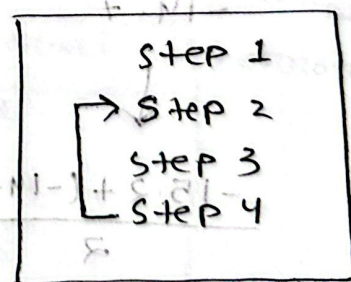
Color not blue

* Prediction (of residual)

by the tree

$$= 71.2 + 0.1 * 16.8 = 72.88$$

Predicted weight = $71.2 + 0.1 * 4.8 = 71.6800$



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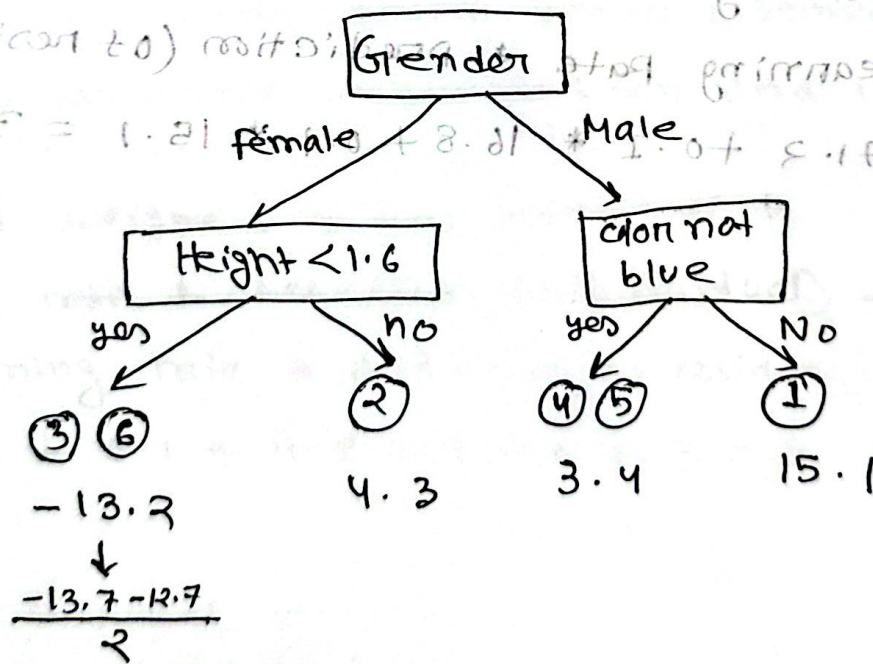
Iteration 2

Again step 2: calc the residuals

Height	color	gender	weight	predicted weight after 1st iteration	residual after 1st iteration
1.8	Blue	Male	88	72.9	15.1
1.6	Green	F	76	71.68	4.3
1.5	Blue	F	56	69.7	-13.7
1.8	Red	M	73	71.6	1.4
1.5	Green	M	77	71.6	5.4
1.4	Blue	M F	57	69.7	-12.7

Step 3 : build a tree to predict residual

new target $\rightarrow$ residual
(coincidentally, same tree)



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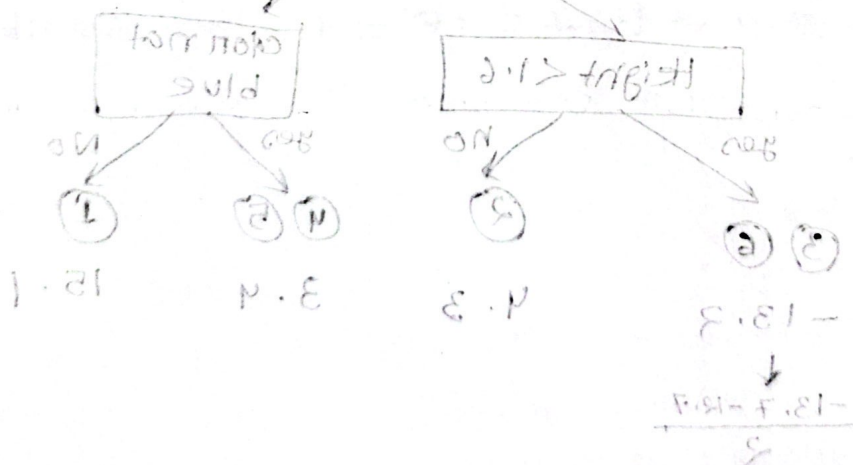
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Step 4: Update the predicted value of the target

Height	color	gender	weight	predicted weight after 2nd iteration
1.6	Blue	Male	88	74.4
1.6	Green	F	76	68.4
1.5	Blue	F	56	68.4
1.8	Red	M	73	71.9
1.5	Green	M	77	71.9
1.4	Blue	F	57	68.4

Predicted weight = initial prediction +

Learning Rate * prediction (of residual) by 1st tree
 + Learning Rate * prediction (of residual) by 2nd tree
 $= 71.2 + 0.1 * 16.8 + 0.1 * 15.1 = 74.39$



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Iteration 3

Height	color	Gender	Weight	Residual after 2nd iteration
1.6	Blue	M	88	13.6
1.6	Green	F	76	3.9
1.6	Blue	F	56	-12.4
1.5	Red	M	73	1.1
1.8	Green	M	77	5.1
1.5	Green	F	57	-11.4
1.4	Blue	M		

will be given how many trees we need to build.

* Given

Gradient boosting - 2 tree = 660 to go

Height = 1.7, color = green, gender = female

calculate predicted weight after 2nd iteration.

predicted weight = initial prediction +

Learning rate * prediction (of residual) by 1st tree

+ Learning rate * prediction (of residual) by 2nd tree

$$= 71.2 + 0.1 * 4.8 + 0.1 * 4.3 = 72.11$$

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Gradient boosting for classification

Loves popcorn	color	age	Loves troll 2 ^{target}	predicted target	Residual ^{step 2}
1.6	Blue	13	yes (1)	0.7	0.3
1.6	Green	87	yes (1)	0.7	0.3
1.6	Blue	44	no (0)	0.7	-0.7
1.5	Blue	44	no (0)	0.7	-0.7
1.8	Red	19	yes (1)	0.7	0.3
1.5	Green	32	yes (1)	0.7	0.3
1.4	Blue	14	yes (1)	0.7	0.3

Step 1: calculate predicted probability of the target column.

$$\text{Log of odds} = \log \frac{\text{No. of 'yes'}}{\text{No. of 'no'}} = \log \left(\frac{P(\text{yes})}{P(\text{no})} \right)$$

$$= \log \left(\frac{4}{2} \right) = 0.7$$

Calculation of $\ln$ in

$$\text{probability of target (Loves troll 2)} = \frac{1}{1 + e^{-0.7}}$$

$$11.54 = 8.14 * 1.0 + 3.4 * 1.0 = 0.7 > 0.5$$

so predicted 1

Step 2: calculate residual between actual target and predicted target.

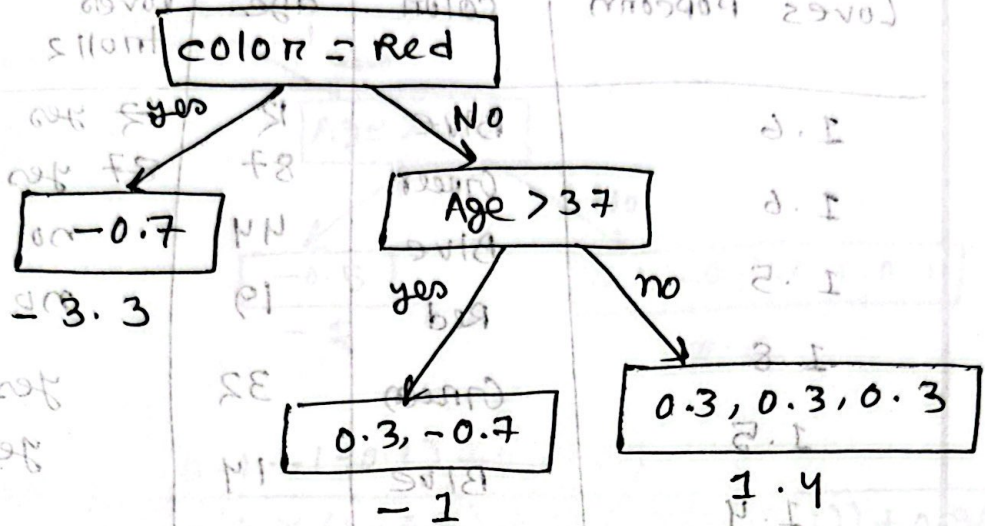
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Step 3: build a tree from the data



value for each leaf = $\frac{\text{sum (residuals)}}{\text{sum} \{ (\text{previous probability}) (1 - \text{previous probability}) \}}$

For the first leaf (color = Red, Yes):

$$= \frac{0.7}{0.7 \times (1 - 0.7)} = -3.3$$

3rd leaf value = $\frac{0.3 + 0.3 + 0.3}{0.7(1 - 0.7) + 0.7(1 - 0.7) + 0.7(1 - 0.7)}$

$= 1.4$

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Step 1: Update the predicted value: 0.9 + 2

Loves popcorn	color	age	loves troll 2	updated predicted target
1.6	Blue	12	yes	0.9
1.6	Green	87	yes	0.5
1.5	Blue	44	no	0.5
1.8	Red	19	no	0.1
1.5	Green	32	yes	0.9
1.4	Blue	14	yes	0.9

Log of odds = initial prediction probability + learning

Rate * prediction (of residual) from the tree

$$= 0.7 + 0.8 \times 1.4 = 1.82$$

$$\text{probability of target (loves troll 2)} = \frac{1}{1 + e^{-1.82}} = 0.9$$

Iteration 2 (step 2)

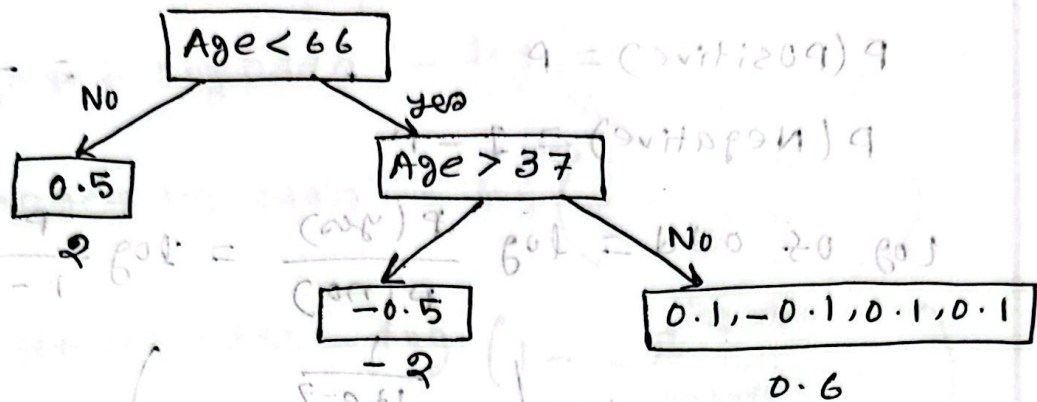
Loves popcorn	color	age	loves troll 2	probability after 1st iteration	Residual
1.6	Blue	12	yes	0.9	0.1
1.6	Green	87	yes	0.5	0.5
1.5	Blue	44	no	0.5	-0.5
1.8	Red	19	no	0.1	-0.1
1.5	Green	32	yes	0.9	0.1
1.4	Blue	14	yes	0.9	0.1

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STEP 3



$$\text{value of leaf} = \frac{0.1 + (1 - 0.1) + 0.1 + 0.1}{0.9 \times (1 - 0.9) + (0.1 \times (1 - 0.1)) + 0.9(1 - 0.9) + 0.9(1 - 0.9)} = 0.6$$

STEP 4

Log of odds = initial prediction probability + learning rate * prediction (of residual) from 1st tree + learning rate * prediction (of residual) from 2nd tree

$$= 0.7 + 0.8 * 1.4 + 0.8 * 0.6 = 2.3$$

$$P(\text{love troll 2}) = \frac{1}{1 + e^{-2.3}} = 0.91$$

Cover Popcorn	color	age	love troll 2	pred after 2nd iteration
1.6	Blue	12	yes	0.91
1.6	green	87	yes	...
1.5	blue	44	yes	...
1.8	Red	19	no	0.15
1.5	green	22	no	...
1.4	blue	14	yes	...

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Derivation

$$P(\text{positive}) = P$$

$$P(\text{Negative}) = 1 - P$$

$$\log \text{ of odd} = \log \frac{P(\text{yes})}{P(\text{no})} = \log \frac{P}{1-P}$$

$$= \log \left(\frac{1+e^{-z}}{1-\frac{1}{1+e^{-z}}} \right)$$

$$= \log \frac{1+e^{-z}}{1+e^{-z}-1}$$

$$= \log \frac{1+e^{-z}}{e^{-z}} = \log \frac{1}{e^{-z}} = \log e^z$$

$$\text{Loss} = -y \log p - (1-y) \log (1-p)$$

$$= -\text{target} \log p - (1-\text{target}) \log (1-p)$$

$$= -\text{target} \log p - \log (1-p) + \text{target} \log (1-p)$$

$$= \text{target} (-\log p + \log (1-p)) - \log (1-p)$$

$$= \text{target} \log \frac{1-p}{p} - \log (1-p)$$

$$= \text{target} \log \left(\frac{p}{1-p} \right) - \log (1-p)$$

$$P = \sigma(z) = \frac{1}{1+e^{-z}} = \frac{e^z}{e^z(1+e^{-z})} = \frac{e^z}{e^z+1}$$

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$$= -1 \text{ target } \log \frac{P}{1-P} - \log(1-P) \quad (1)$$

$$= -1 \text{ target } \log \text{ odds} - \log \left(1 - \frac{1}{1+e^z} \right)$$

$$= -1 \text{ target } \log \text{ odds} - \log \left(1 - \frac{e^z}{e^z + 1} \right)$$

$$= -1 \text{ target } \log \text{ odds} - \log \left(1 - \frac{e^{\log \text{ odds}}}{e^{\log \text{ odds}} + 1} \right)$$

$$= -1 \text{ target } \log \text{ odds} - \log \left(\frac{e^{\log \text{ odds}} + 1 - e^{\log \text{ odds}}}{e^{\log \text{ odds}} + 1} \right)$$

$$= -1 \text{ target } \log \text{ odds} - \log \frac{1}{e^{\log \text{ odds}} + 1}$$

$$= -1 \text{ target } \log \text{ odds} - \log (e^{\log \text{ odds}} + 1)$$

$$\text{Loss} = -1 \text{ target } \log \text{ odds} + \log (e^{\log \text{ odds}} + 1)$$

$$\frac{\partial \text{Loss}}{\partial \log \text{ odds}} = -1 \text{ target} + \frac{1}{e^{\log \text{ odds}} + 1} \times e^{\log \text{ odds}}$$

$$= \sum_{i=1}^n -1 \text{ target}_i + P$$

$$-1 + P - 1 + P - 0 + P - 0 + P - 1 + P - 1 + P = 0$$

$$\Rightarrow -4 + 6P = 0 \therefore P = 4/6$$

$$1 - P = 1 - 4/6 = 2/6$$

$$\log \text{ odds} = \log \frac{P}{1-P} = \frac{4/6}{2/6} = 4/2 = 2$$

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